

ECON484 Machine Learning
Midterm Exam
04.05.2026 - KEY

Student Number		Question	Points
Name & Surname		1	___ / 5
Instructions: - This exam is in two parts. The first part of the exam is closed resource. The questions in this part are really simple and you should not need any assistance in that part. The second part of the exam is open resource. A resource could be anything from your lecture notes to Artificial Intelligence tools. However, your classmates and other persons are not resources. - In the first part you will have some simple calculated results by hand, and in the second part you will have to reflect on those results. You may be making some mistakes in the first part. You may even be expected to have mistakes. This is by design, so that you can reflect on those mistakes in the second part. Do not spend excessive time to re-check Part 1. Total exam duration is 75 minutes . You are not allowed to use your resources in the first 30 minutes to cover Part 1.		2	___ / 20
		3	___ / 25
		4	___ / 15
		5	___ / 10
		6	___ / 15
		7	___ / 10
		Total	___ / 100

PART 1 – CLOSED RESOURCE (50 Points)

QUESTION 1 (Seeds, Random Numbers and Shuffling; 5 points)

In this question you will create your unique random number seed that you will use in this exam. **Based on the seed, you will have different computational results than your classmates.**

Follow this procedure:

- First add the digits on you student number (ie. 1234567 means $1+2+3+4+5+6+7=28$)
- Then find the remainder of this number when divided by 5 (ie. 28 divided by 5 is 5 with remainder 3)
- The seed will be this remainder. (ie. seed is 3)

Calculations:

Seed: 3

1234567 $\rightarrow$ $1+2+3+4+5+6+7=28$
28 / 5 gives remainder as 3

Now you can read the case study.

CASE STUDY

You were in your favorite coffee shop with your classmate from ECON484 and discussing experiment ledgers and how they could be used when trying to predict popular indexes such as stock market or USD. Then someone interrupts you: An older man with a grey goatee, dark circles around his eyes an wide smile.

He introduces himself. Apparently he was a big time stock broker back in the 80s and 90s when the markets were roaring. He says their generation never applied complex algorithms, or methodical work like they do now. Nevertheless their generation made tenfold more money. His view is that the more complex things get, the more mistakes are made.

You like his approach and ask him, how he would track USD back in the day. He says he had a simple average-based estimation algorithm, and based on the algorithm he would not decide on the exact amount, but the up or down behavior of the value. This makes some sense into you. Most modern estimation algorithms are modified versions of a moving average anyway. The idea of up versus down also makes some sense. Most investors would not bother themselves with how much as long as they are in the winning side. So you ask more about this algorithm.

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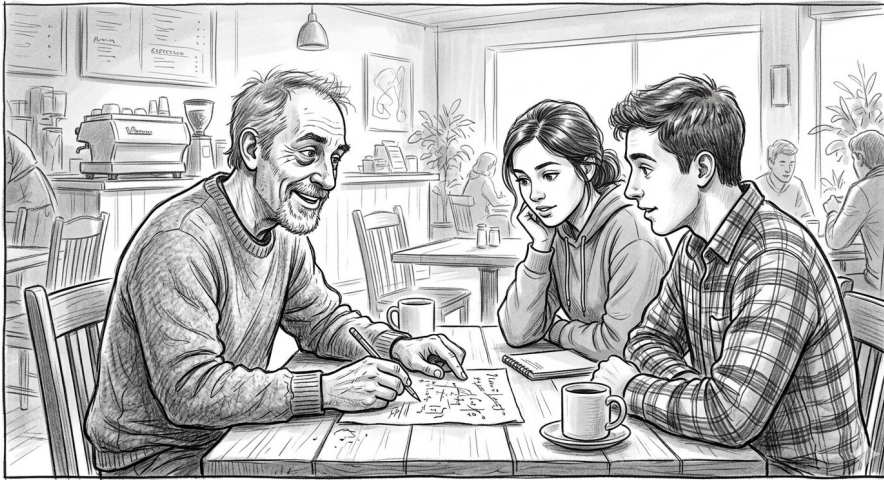


Figure 1: You listening to a winning strategy from the 90s.

His algorithm is as follows:

1. Keep the past 25 trading days data at hand. I would have the actual value, and if it was “up” or “down”
2. Pick 5 random days from this data. I used to shuffle the data set, and pick the last 5 days in the shuffled deck, but you could pick a different approach.
3. Those 5 random days have a vote (ie. 3 ups and 2 downs means an “up”)
4. You select the result of the vote as your position.

Because you are a Gen-Z you do not take his word for granted and like to take this algorithm for a spin. You decide to:

1. Collect USD Forex data for a period of 25 business days and construct a table
2. Use your own seed to shuffle the data set as instructed and pick the five days
3. Estimate whether it is up or down in the next business day.

Day	March 2	March 3	March 4	March 5	March 6
USD	43,8850	43,8913	43,8932	43,9135	43,9055
Vote	UP	UP	UP	UP	DOWN
Day	March 9	March 10	March 11	March 12	March 13
USD	44,0007	43,9826	43,9964	44,0310	44,0210
Day	UP	DOWN	DOWN	UP	DOWN
Day	March 16	March 17	March 18	March 24	March 25
USD	44,1060	44,1207	44,1325	44,2636	44,2737
Vote	UP	UP	UP	UP	UP
Day	March 26	March 27	March 30	March 31	April 1
USD	44,2828	44,2887	44,3841	44,3961	44,3938
Vote	UP	UP	UP	UP	DOWN
Day	April 2	April 3	April 6	April 7	April 8
USD	44,4155	44,4306	44,5085	44,5255	44,4527
Vote	UP	DOWN	UP	UP	DOWN

Table 1. USD Forex Values and Their “Votes”

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You also note that **on April 9th, the USD Forex Buying value was 44,5102 which means an “up” ticker.**

Then you ask your favorite AI agent to create a simple Python script to shuffle the dates as instructed and the fun begins.

QUESTION 2 (Using Shuffled Results; 20 points)

When you use your AI assistant to create some code, you also **execute the code** and get results (you can check the appendix if you'd like, but no need for this question). Here are the results, in a nice tabular form for you.

Seed	Shuffled List of Days
0	['March 25', 'March 2', 'April 7', 'March 31', 'March 10', 'March 9', 'March 11', 'March 16', 'April 6', 'April 2', 'March 6', 'March 4', 'April 3', 'March 5', 'March 30', 'March 17', 'April 1', 'March 13', 'April 8', 'March 26', 'March 27', 'March 12', 'March 3', 'March 24', 'March 18']
1	['March 16', 'April 2', 'April 8', 'April 3', 'April 7', 'March 9', 'April 6', 'March 24', 'March 27', 'March 13', 'March 17', 'March 30', 'March 2', 'March 11', 'March 3', 'March 10', 'March 18', 'April 1', 'March 25', 'March 26', 'March 5', 'March 12', 'March 4', 'March 31', 'March 6']
2	['March 5', 'April 1', 'March 27', 'March 18', 'March 2', 'April 6', 'March 6', 'March 11', 'March 24', 'March 26', 'March 31', 'April 3', 'March 25', 'March 16', 'March 30', 'April 2', 'April 8', 'March 10', 'March 12', 'March 13', 'March 9', 'March 17', 'April 7', 'March 4', 'March 3']
3	['March 10', 'March 27', 'April 6', 'March 25', 'March 3', 'March 9', 'April 2', 'March 16', 'March 13', 'March 24', 'April 8', 'March 18', 'March 5', 'March 12', 'April 3', 'April 7', 'March 2', 'March 4', 'March 26', 'April 1', 'March 17', 'March 6', 'March 30', 'March 31', 'March 11']
4	['April 8', 'March 16', 'March 24', 'March 10', 'March 17', 'March 3', 'March 30', 'March 9', 'April 7', 'March 25', 'April 6', 'March 27', 'April 2', 'April 1', 'March 12', 'April 3', 'March 2', 'March 31', 'March 4', 'March 6', 'March 26', 'March 18', 'March 5', 'March 13', 'March 11']

Table 2. Seed-Based Shuffle Results

a) (10 points) Based on your **personal unique seed** in Question 1, **select your shuffle results (ie. circle a seed in Table 2) and then pick the last 5 days from Table 2.** Then based on your selected five days from Table 2, **fill out the following table using data from Table 1.**

	Day 1	Day 2	Day 3	Day 4	Day 5
Actual Day	March 17	March 6	March 30	March 31	March 11
USD Value	44,1207	43,9055	44,3841	44,3961	43,9964
Vote	UP	DOWN	UP	UP	DOWN

Table 3. Voting for April 9th

b) (5 points) Now hold the vote based on Table 3 and predict if it is an “up” or “down” for April 9th.

3 UP and 2 DOWN votes. Result is UP. My prediction is an UP ticker.

c) (5 points) Does your prediction hold?

This prediction holds.

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QUESTION 3 (Ledgers, Understanding ML Process, Reproducibility, Metrics; 25 points)

In this question you are given very simple and straightforward questions that ask you to translate this case into machine learning terminology (ie. Talk the talk).

a) (5 points) You ask yourself and your classmate about use of ledgers for this algorithm. What type of data would you need to store in the ledger for this type of up/down estimation?

The ledger should store historical training data – list of dates used in training, USD forex buying values, and binary labels (Up/Down). For reproducibility, it must also record sampling metadata, including the random seed and the indices of the five selected days.

b) (5 points) If this is an ML based algorithm, what would be the “features” in this algorithm?

The "features" are the inputs the model uses to make a guess. In this case, the features are the up/down votes of the 5 random days I picked from Table 1.

c) (5 points) If this is an ML based algorithm, what would be the “output” in this algorithm?

The "output" is the final answer the model gives you. For this exam, the output is a single prediction for April 9th: either "UP" or "DOWN".

d) (5 points) If your classmate uses the same student number and the same Table 2 as you, will they get a different result? Why or why not?

No, they will not get a different result. Even though the process is called "random," it follows a strict mathematical rule set by the seed. If two people use the same student number, they get the same seed. If they use the same seed on the same list of dates, the computer will always shuffle the list in the exact same order. Therefore, their "last 5 days" will be identical to mine.

e) (5 points) Looking at Table 1 what is the frequency of “up” and “down” votes in the 25 day sample? Then calculate the frequency of “up” and “down” votes, but only for the days where the USD value was above 44,10000. Please show your work in detail (ie. which days are selected). How does this selective frequency compare to the overall sample frequency? What does this tell you about the model's potential bias?

For the 25 days we have 18 UP and 7 DOWN votes. Which means 72 percent have UP votes.

There are 15 days with USD above 44,100: March 16 (UP), 17 (UP), 18 (UP), 24 (UP), 25 (UP), 26 (UP), 27 (UP), 30 (UP), 31 (UP), April 1 (DOWN), 2 (UP), 3 (DOWN), 6 (UP), 7 (UP), 8 (DOWN). Out of these we have 12 UP votes and 3 DOWN votes. This is 80 percent UP.

The increase from 72% to 80% shows that as the price stays high, the algorithm becomes even more likely to predict an "UP" movement. This indicates a strong momentum bias (trend following), where the model assumes that because things have been going up, they will continue to go up. The model ignores the possibility that a high price might be "overbought" or due for a correction. By relying on a simple vote of past days, it cannot detect when a trend is losing steam or about to reverse. In ML terms, this training set has a class imbalance (more UP than DOWN). A simple majority-vote model will naturally be biased toward the majority class (UP), making it "blind" to the signals that precede a DOWN ticker.

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PART 2 – OPEN RESOURCE (50 Points)

In Part 2, you may use external resources, including AI tools. However:

- The questions are intentionally structured so that generic prompts (e.g., “give me the correct answer”) will not be sufficient.
- You are responsible for verifying and understanding any output you use.
- Answers will be graded primarily on reasoning and clarity, not just the final result.
- If you use AI, briefly indicate how it assisted you (e.g., idea generation, structuring, verification).

Submissions that demonstrate clear understanding will receive full credit, even if intermediate results from Part 1 contain errors.

QUESTION 4 (Assessment of Model Fragility; 15 points)

a) (10 points) This model predicts a single day, however you could easily back-test for a longer period. Describe the procedure for back-testing for the whole month of April. You should still stick to the random shuffling concept in your procedure.

We should define the target window (list of days to be predicted for April), and for each day determine the training window (25 days prior) so that the training data set can be created. This means we will have a separate data set for each prediction. For every target day, we will apply the unique seed to shuffle that specific 25 day dataset and pick the last 5 days from the shuffled deck, and hold a vote. In a ledger we will record model's prediction versus the actual market outcome. Then based on the test we will calculate the accuracy score for each day and also record on the ledger. We may be dependent on a particular seed, so this procedure should also be run with different seeds.

b) (5 points) Eventually the algorithm would fail on some days. For those days, does the algorithm fail because of the logic, or because the sample size ($n=5$) is too small to capture market volatility?

The primary failure is the logic. The model assumes the market is a closed system where 5 random days from the past can represent the future, ignoring external economic shocks, news, or trend reversals. It treats trading days as independent events that can be shuffled. It borrows concepts from classification and clustering, but as there is a time series here, it has to be handled with different assumptions. The small sample size ($n=5$) is also problematic but it only adds to the logic error. Even if the logic was solid, using only 5 votes out of 25, the model would be highly sensitive.

QUESTION 5 (Drift; 10 points)

a) (5 points) Calculate the average USD value for the first 5 days and the last 5 days of Table 1. Use these two specific averages to justify whether the Broker's 90s algorithm is experiencing Data Drift in 2026. A generic answer without these two calculated values will receive zero credit.

First 5 days: $(43.8850+43.8913+43.8932+43.9135+43.9055)/5=43.8977$.

Last 5 days: $(44.4155+44.4306+44.5085+44.5255+44.4527)/5=44.46656$.

The statistical distribution of the input data has changed significantly between the start and end of the sample period. The algorithm is experiencing Data Drift. Randomness in the algorithm gives a false sense of security against drift. But it enforces bias towards the past.

b) (5 points) The Broker claims this worked in the 90s. Based on the 2026 data in Table 1, identify signs of Concept Drift that make this “simple” strategy dangerous today.

Concept Drift occurs when the relationship between the input (past votes) and the output (tomorrow's movement) changes. The data shows that even when the USD value is high and most votes are "UP" (80% frequency), the market still produces "DOWN" tickers (e.g., April 8th). So we cannot immediately assume a relationship between trend and result. On the other hand, the model's 72–80% bias toward "UP" makes it blind to shifts or corrections.

QUESTION 6 (Data Leakage and Validation; 15 points)

a) (5 points) Explain if and why the Broker's "random shuffling" method might lead to Data Leakage.

In time-series forecasting, the golden rule is that the future must never inform the past.

Regarding the Broker's shuffling; as long as sampling within the training data is independent of the target value and is done per-instance (ie. we can replicate the procedure) it should be generally safe, and would not include direct leakage. In other words, direct leakage would be introduced only if the 5 days are selected using global dataset properties (ie. pick most volatile days, etc). In this approach we use random selection, so we are not tied to dataset properties. The sample is a uniform subset of the feature window (up/down values for 25 days).

The approach has other problems, in particular to bias and stationarity assumption but not that of leakage.

b) (10 points) Propose a walk-forward validation structure using the dates in Table 1 that would be more scientifically sound. If needed you can also draw assistive diagrams to describe how you handle the data set.

Instead of treating the 25-day window as a static bag of "shuffled" votes, we must implement a Walk-Forward Validation (time-series cross-validation) structure. This approach directly addresses the stationarity assumption and recency bias that the Broker's algorithm ignores.

I would revise the procedure like this:

- Use Sequential Windowing (No Shuffling): Shuffling creates a distributional bias. Abandon the "shuffled deck" concept. Maintain the natural chronological order of the days to preserve the autocorrelation of the price movements.
- Use Weighted Feature Sets: Rather than picking 5 days at random, use a sliding window of the most recent n days (e.g., the last 5/10/25 trading days) to make the prediction. This acknowledges that the market is non-stationary—the relationship between "Up/Down" votes on March 2nd is likely irrelevant to the volatility of April 9th.
- Continue to use voting.

Validation folds would work like this in order to have a chance to detect trend reversal:

- Fold 1 – Train on $n=5$ days (March 25th through to March 31st) and estimate April 1st.
- Fold 2 – Train on $n=5$ days (March 26th through April 1st) and estimate April 2nd.

This model responds to clusters of volatility. However, the parameter $n=5$ is not necessarily arbitrarily selected. It should be treated as a hyperparameter, and be tuned/optimized through more validation runs.

QUESTION 7 (Short Reflection Questions; 10 points)

a) (5 points) The "Up/Down" vote is a primitive form of data encoding. Turning continuous USD values into binary votes loses information. **What specific information are we losing** by encoding the data this way, and **how would a weighting feature change the model's sensitivity?**

Information Loss: We lose magnitude and velocity. A binary UP vote treats a +0.0001 change (noise) exactly the same as a +10.000 change (significant breakout). We also lose the ordinal relationship between days—the distance between prices, and therefore the velocity of change.

Weighting Feature: Introducing a weighting feature would prioritize more recent votes over older ones. With the encoding of up/down, Since the data is encoded as primitive binary UP/DOWN votes, traditional mathematical weighting like exponential decay which works on continuous values are difficult to apply directly to the labels. Sliding window subsets (ie. using the most recent days' votes only) or frequency weighting (ie. giving multiple votes to more recent dates) would improve the system.

b) (5 points) If you developed Python code to execute this algorithm (with or without AI assistance), and you wanted an automated process with the program automatically writing into the ledger, you'd have to take into account ledger specific issues as well. AI assistance in this regard gives a standard list: seeds, versioning, timestamps. Let's **introduce a constraint**. Assume your ledger has **a storage limit of only 3 fields per entry**, and you are designing the ledger part of shuffling (ie. Appendix 1). Which 3 specific pieces of metadata are mechanically mandatory to reproduce the shuffle results in Table 2? Justify your decision.

To achieve 100% reproducibility in a constrained ledger, the following three fields are mandatory:

1. **The seed value – The seed we use to set the random number generator**
2. **The input list state (Initial Order) – This is the original order of the data**
3. **The algorithm/library version – Assuming we have Python, another computer could have a different version or even a different library.**

Appendix 1. Sample Python code to shuffle the dates.

```
import random

# The list of dates provided
dates = [
    "March 2", "March 3", "March 4", "March 5", "March 6",
    "March 9", "March 10", "March 11", "March 12", "March 13",
    "March 16", "March 17", "March 18", "March 24", "March 25",
    "March 26", "March 27", "March 30", "March 31", "April 1",
    "April 2", "April 3", "April 6", "April 7", "April 8"
]

seeds = [0, 1, 2, 3, 4]

for seed in seeds:
    # Work on a copy to keep the original list intact for each iteration
    shuffled_dates = dates.copy()

    # Initialize the random number generator with the specific seed
    random.seed(seed)

    # Perform the in-place shuffle
    random.shuffle(shuffled_dates)

    print(f"--- Seed {seed} ---")
    print(shuffled_dates)
    print("\n")
```